INMO(20^{th}) – 2005

Time – 3 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
- Ruler and compass allowed.
- Answer all the questions.

1. Let M be the midpoint of side BC of a triangle ABC. Let the median AM intersect the incircle of ABC at K and L, K being nearer to A than L. If AK = KL = LM. Prove that the sides of triangle ABC are in the ration 5 : 10 : 13 in some order.
2. Let α and β be positive integers such that $\frac{43}{197} < \frac{\alpha}{\beta} < \frac{17}{77}$. Find the minimum possible value of β.
3. Let p, q, r be positive real numbers, not all equal, such that some two of the equations $px^2 + 2qx + r = 0$, $qx^2 + 2rx + p = 0$, $rx^2 + 2px + q = 0$ have a common root, say α. Prove that:
   (a) α is real and negative.
   (b) The remaining third equation has non real roots.
4. All possible 6-digit numbers, in each of which the digits occur in no-increasing order (from left to right, e.g. 877550) are written as a sequence in increasing order. Find the 2005th number in this sequence.
5. Let x_1 be a given positive integer. A sequence $(x_n)_{n=1}^{\infty} = (x_1, x_2, x_3, ...)$ of positive integers such that x_n for $n \geq 2$, is obtained from x_{n-1} by adding some nonzero digit of x_{n-1}. Prove that:
   (a) The sequence has an even number.
   (b) The sequence has infinitely many even numbers.
6. Find all functions $f : \Re \to \Re$ such that $f(x^2 + yf(z)) = xf(x) + zf(y)$ for all $x, y, z \in \Re$, where $\Re$ denotes the set of all real numbers.

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